**1)**

$\frac{2}{x(x^2+2)} \equiv \frac{A}{x} + \frac{Bx+C}{x^2+2}$

$2 = A(x^2 + 2) + (Bx + C)x$

Let $x = 0$: $\quad 2 = 2A \therefore A = 1$

Let $x = 1$: $\quad 2 = 3A + B + C$ so $B + C = -1$

Let $x = 2$: $\quad 2 = 6A + 4B + 2C$ so $4B + 2C = -4 \therefore B = -1$ and $C = 0$

$\frac{2}{x(x^2+2)} = \frac{1}{x} - \frac{x}{x^2+2}$

**2)**

$\frac{3x^2-x+10}{(x+1)(x^2+1)} \equiv \frac{A}{x+1} + \frac{Bx+C}{x^2+1}$

$3x^2 - x + 10 = A(x^2 + 1) + (Bx + C)(x + 1)$

Let $x = 0$: $\quad 10 = A + C$

Let $x = 1$: $\quad 12 = 2A + 2B + 2C$ so $6 = A + B + C \ \therefore B = -4$

Let $x = 2$: $\quad 20 = 5A + 6B + 3C \therefore A = 7, C = 3$

$\frac{3x^2-x+10}{(x+1)(x^2+1)} = \frac{7}{x+1} - \frac{4x-3}{x^2+1}$

**3) a)**

The term in the numerator of the fraction with denominator or $x^2 + 2$ should be of the form $Bx + C$

**3) b)**

$\frac{3x^2+5x+5}{(x+1)(x^2+2)} \equiv \frac{A}{x+1} + \frac{Bx+C}{x^2+2}$

$3x^2 + 5x + 5 = A(x^2 + 2) + (Bx + C)(x + 1)$

Let $x = 0$: $\quad 5 = 2A + C$

Let $x = 1$: $\quad 13 = 3A + 2B + 2C$

Let $x = 2$: $\quad 27 = 6A + 6B + 3C$

$\therefore A = 1, B = 2$ and $C = 3$

$\frac{3x^2+5x+5}{(x+1)(x^2+2)} = \frac{1}{x+1} + \frac{2x+3}{x^2+2}$

**4) a)**

$$\frac{x+5}{(x^2+1)(x+1)} \equiv \frac{A}{x+1} + \frac{Bx+C}{x^2+1}$$

$x + 5 = A(x^2 + 1) + (Bx + C)(x + 1)$

Let $x = 0$: $\quad 5 = A + C$

Let $x = 1$: $\quad 6 = 2A + 2B + 2C$

Let $x = 2$: $\quad 7 = 5A + 6B + 3C$

$\therefore A = 2, B = -2$ and $C = 3$

$$\frac{x+5}{(x^2+1)(x+1)} = \frac{2}{x+1} + \frac{-2x+3}{x^2+1} = \frac{2}{x+1} - \frac{2x-3}{x^2+1}$$

**4) b)**

$$\frac{5x}{(2x+1)(x^2+1)} \equiv \frac{A}{2x+1} + \frac{Bx+C}{x^2+1}$$

$5x = A(x^2 + 1) + (Bx + C)(2x + 1)$

Let $x = 0$: $\quad 0 = A + C$

Let $x = 1$: $\quad 5 = 2A + 3B + 3C$

Let $x = 2$: $\quad 10 = 5A + 10B + 5C$

$\therefore A = -2, B = 1$ and $C = 2$

$$\frac{5x}{(2x+1)(x^2+1)} = -\frac{2}{2x+1} + \frac{x+2}{x^2+1}$$

**4) c)**

$$\frac{x-2}{(x+3)(x^2+4)} \equiv \frac{A}{x+3} + \frac{Bx+C}{x^2+4}$$

$x - 2 = A(x^2 + 4) + (Bx + C)(x + 3)$

Let $x = 0$: $\quad -2 = 4A + 3C$

Let $x = 1$: $\quad -1 = 5A + 4B + 4C$

Let $x = 2$: $\quad 0 = 8A + 10B + 5C$

$\therefore A = -\frac{5}{13}, B = \frac{5}{13}$ and $C = -\frac{2}{13}$

$$\frac{x-2}{(x+3)(x^2+4)} \equiv -\frac{5}{13(x+3)} + \frac{5x-2}{13(x^2+4)}$$

**4) d)**

$\frac{3x^2-6x+15}{(2x^2+1)(x-1)} \equiv \frac{A}{x-1} + \frac{Bx+C}{2x^2+1}$

$3x^2 - 6x + 15 = A(2x^2 + 1) + (Bx + C)(x - 1)$

Let $x = 0$: $\quad 15 = A - C$

Let $x = 1$: $\quad 12 = 3A \ \therefore A = 4$ and $C = -11$

Let $x = 2$: $\quad 15 = 9A + 2B + C$

$\therefore A = 4, B = -5$ and $C = -11$

$\frac{3x^2-6x+15}{(2x^2+1)(x-1)} = \frac{4}{x-1} - \frac{5x+11}{2x^2+1}$

**4) e)**

Let f$(x) = 2x^3 - 3x^2 + 8x - 12$ so f$\left(\frac{3}{2}\right) = \frac{27}{4} - \frac{27}{4} + 12 - 12 = 0 \therefore (2x - 3)$ is a factor.

$2x^3 - 3x^2 + 8x - 12 = (2x - 3)(x^2 + 4)$

$\frac{25x^2}{2x^3-3x^2+8x-12} = \frac{25x^2}{(2x-3)(x^2+4)} \equiv \frac{A}{2x-3} + \frac{Bx+C}{x^2+4}$

$25x^2 = A(x^2 + 4) + (Bx + C)(2x - 3)$

Let $x = 0$: $\quad 0 = 4A - 3C$

Let $x = 1$: $\quad 25 = 5A - B - C$

Let $x = 2$: $\quad 100 = 8A + 2B + C$

$\therefore A = 9, B = 8$ and $C = 12$

$\frac{25x^2}{2x^3-3x^2+8x-12} = \frac{9}{2x-3} + \frac{8x+12}{x^2+4}$

**4) f)**

Let $f(x) = x^3 - 3x^2 + 5x - 15$ so $f(3) = 27 - 27 + 15 - 15 = 0 \therefore (x-3)$ is a factor.

$x^3 - 3x^2 + 5x - 15 = (x-3)(x^2+5)$

$\frac{x^2}{x^3-3x^2+5x-15} = \frac{x^2}{(x-3)(x^2+5)} \equiv \frac{A}{x-3} + \frac{Bx+C}{x^2+5}$

$x^2 = A(x^2+5) + (Bx+C)(x-3)$

Let $x = 0$: $\quad 0 = 5A - 3C$

Let $x = 1$: $\quad 1 = 6A - 2B - 2C$

Let $x = 2$: $\quad 4 = 9A - 2B - C$

$\therefore A = \frac{9}{14}, B = \frac{5}{14}$ and $C = \frac{15}{14}$

$\frac{x^2}{x^3-3x^2+5x-15} = \frac{9}{14(x-3)} + \frac{5x+15}{14(x^2+5)}$

**4) g)**

$\frac{1}{(x^2+k)(x-1)} \equiv \frac{A}{x-1} + \frac{Bx+C}{x^2+k}$

$1 = A(x^2+k) + (Bx+C)(x-1)$

Let $x = 0$: $\quad 1 = kA - C$

Let $x = 1$: $\quad 1 = (k+1)A \;\; \therefore A = \frac{1}{k+1}$ and $C = \frac{k}{k+1} - 1 = -\frac{1}{k+1}$

Compare squared terms: $0 = A + B$

$\therefore A = \frac{1}{k+1}, B = -\frac{1}{k+1}$ and $C = -\frac{1}{k+1}$

$\frac{1}{(x^2+k)(x-1)} = \frac{1}{(k+1)(x-1)} - \frac{x+1}{(k+1)(x^2+k)}$

**4) f)**

$\frac{k}{(x^2+k)(x-k)} \equiv \frac{A}{x-k} + \frac{Bx+C}{x^2+k}$

$1 = A(x^2+k) + (Bx+C)(x-k)$

Let $x = 0$: $\quad 1 = kA - kC$

Let $x = k$: $\quad 1 = (k^2+k)A \therefore A = \frac{1}{k+1}$ and $C = \frac{k\left(\frac{1}{k+1}\right)-1}{k} = -\frac{k}{k+1}$

Compare squared terms: $0 = A + B$

$\therefore A = \frac{1}{k+1}, B = -\frac{1}{k+1}$ and $C = -\frac{k}{k+1}$

$\frac{k}{(x^2+k)(x-k)} = \frac{1}{(k+1)(x-k)} - \frac{x+k}{(k+1)(x^2+k)}$

**5)**

$\frac{6x^2+5x+6}{x^3+kx} \equiv \frac{A}{x} + \frac{Bx+C}{x^2+k}$

$6x^2 + 5x + 6 = A(x^2 + k) + (Bx + C)x$

Let $x = 0$: $\quad 6 = kA \therefore A = \frac{6}{k}$

So since $A$ is a positive integer $k = 1$, $k = 2$, $k = 3$ or $k = 6$

Compare quadratic terms: $6 = A + B \Rightarrow 6 = \frac{6}{k} + B \therefore B = 6 - \frac{6}{k}$

So since $B$ is a positive integer $k = 2$, $k = 3$ or $k = 6$

Compare linear terms: $5 = C$

$\therefore k = 2$, $k = 3$ or $k = 6$

**6) a)**

$\frac{18}{(x-1)(x+2)(x-4)} \equiv \frac{A}{x-1} + \frac{B}{x+2} + \frac{C}{x-4}$

$18 = A(x + 2)(x - 4) + B(x - 1)(x - 4) + C(x - 1)(x + 2)$

Let $x = 1$: $\quad 18 = -9A \therefore A = -2$

Let $x = -2$: $18 = 18B \ \therefore B = 1$

Let $x = 4$: $\quad 18 = 18C \ \therefore C = 1$

$\frac{18}{(x-1)(x+2)(x-4)} = -\frac{2}{x-1} + \frac{1}{x+2} + \frac{1}{x-4}$

So that $\int \frac{18}{(x-1)(x+2)(x-4)}\,\mathrm{d}x = \int \left(-\frac{2}{x-1} + \frac{1}{x+2} + \frac{1}{x-4}\right)\mathrm{d}x$

$= -2\ln|x - 1| + \ln|x + 2| + \ln|x - 4| + c$

$= -\ln|(x - 1)^2| + \ln|(x + 2)(x - 4)| + c$

$= \ln\left|\frac{(x+2)(x-4)}{(x-1)^2}\right| + c$

**6) b)**

$\frac{x-2}{(x-1)^2} \equiv \frac{A}{x-1} + \frac{B}{(x-1)^2}$

$x - 2 = A(x-1) + B$

Let $x = 1$: $\quad -1 = B \therefore B = -1$

Let $x = 0$: $\quad -2 = -A + B \ \therefore A = 1$

$\frac{x-2}{(x-1)^2} = \frac{1}{x-1} - \frac{1}{(x-1)^2}$

So that $\int \frac{x-2}{(x-1)^2}\,\mathrm{d}x = \int \left(\frac{1}{x-1} - \frac{1}{(x-1)^2}\right)\mathrm{d}x = \int \left(\frac{1}{x-1} - (x-1)^{-2}\right)\mathrm{d}x$

$= \ln|x-1| + (x-1)^{-1} + c$

$= \ln|x-1| + \frac{1}{x-1} + c$

**6) c)**

$\frac{7x}{(2x-1)(x+3)^2} \equiv \frac{A}{2x-1} + \frac{B}{x+3} + \frac{C}{(x+3)^2}$

$7x = A(x+3)^2 + B(2x-1)(x+3) + C(2x-1)$

Let $x = \frac{1}{2}$: $\quad \frac{7}{2} = \frac{49}{4}A \therefore A = \frac{2}{7}$

Let $x = -3$: $-21 = -7C \ \therefore C = 3$

Let $x = 0$: $\quad 0 = 9A - 3B - C \therefore B = -\frac{1}{7}$

$\frac{7x}{(2x-1)(x+3)^2} \equiv \frac{2}{7(2x-1)} - \frac{1}{7(x+3)} + \frac{3}{(x+3)^2}$

So that $\int \frac{7x}{(2x-1)(x+3)^2}\,\mathrm{d}x = \int \left(\frac{2}{7(2x-1)} - \frac{1}{7(x+3)} + \frac{3}{(x+3)^2}\right)\mathrm{d}x$

$= \int \left(\frac{2}{7(2x-1)} - \frac{1}{7(x+3)} + 3(x+3)^{-2}\right)\mathrm{d}x$

$= \frac{1}{7}\ln|2x-1| - \frac{1}{7}\ln|x+3| - 3(x+3)^{-1} + c$

**7)**

$\frac{3x^2+x+4}{(x+1)(x^2+2)} \equiv \frac{A}{x+1} + \frac{Bx+C}{x^2+2}$

$3x^2 + x + 4 = A(x^2 + 2) + (Bx + C)(x + 1)$

Let $x = 0$: $4 = 2A + C$

Let $x = 1$: $8 = 3A + 2B + 2C$

Let $x = 2$: $18 = 6A + 6B + 3C$

$\therefore A = 2, B = 1$ and $C = 0$

$\frac{3x^2+x+4}{(x+1)(x^2+2)} = \frac{2}{x+1} + \frac{x}{x^2+2}$

So that $\int \frac{3x^2+x+4}{(x+1)(x^2+2)}\,\mathrm{d}x = \int \left(\frac{2}{x+1} + \frac{x}{x^2+2}\right)\mathrm{d}x$

$= 2\ln|x+1| + \frac{1}{2}\ln|x^2+2| + c$

$= \ln|(x+1)^2| + \ln\left|\sqrt{x^2+2}\right| + c$

$= \ln\left|(x+1)^2\sqrt{x^2+2}\right| + c$ as req.

**8)**

$\frac{x^2-4x+15}{(x+1)(x^2+3)} \equiv \frac{A}{x+1} + \frac{Bx+C}{x^2+3}$

$x^2 - 4x + 15 = A(x^2 + 3) + (Bx + C)(x + 1)$

Let $x = 0$: $15 = 3A + C$

Let $x = 1$: $12 = 4A + 2B + 2C$

Let $x = 2$: $11 = 7A + 6B + 3C$

$\therefore A = 5, B = -4$ and $C = 0$

$\frac{x^2-4x+15}{(x+1)(x^2+3)} = \frac{5}{x+1} - \frac{4x}{x^2+3}$

So that $\int \frac{x^2-4x+15}{(x+1)(x^2+3)}\,\mathrm{d}x = \int \left(\frac{5}{x+1} - \frac{4x}{x^2+3}\right)\mathrm{d}x$

$\therefore \int_1^4 \frac{x^2-4x+15}{(x+1)(x^2+3)}\mathrm{d}x = \int_1^4 \left(\frac{5}{x+1} - \frac{4x}{x^2+3}\right)\mathrm{d}x = [5\ln|x+1| - 2\ln|x^2+3|]_1^4$

$= (5\ln 5 - 2\ln 19) - (5\ln 2 - 2\ln 4)$

$= \ln\left(\frac{5^5}{19^2}\right) - \ln\left(\frac{2^5}{4^2}\right)$

$= \ln\left(\frac{3125}{361}\right) - \ln(2)$

$= \ln\left(\frac{3125}{722}\right)$ as req.

**9)**

$\frac{4x^2+6x+2}{(x+2)(x^2+2)} \equiv \frac{A}{x+2} + \frac{Bx+C}{x^2+2}$

$4x^2 + 6x + 2 = A(x^2 + 2) + (Bx + C)(x + 2)$

Let $x = 0$: $2 = 2A + 2C$

Let $x = 1$: $12 = 3A + 3B + 3C$

Let $x = 2$: $30 = 6A + 8B + 4C$

$\therefore A = 1, B = 3$ and $C = 0$

$\frac{4x^2+6x+2}{(x+2)(x^2+2)} = \frac{1}{x+2} + \frac{3x}{x^2+2}$

So that $\int \frac{4x^2+6x+2}{(x+2)(x^2+2)}\,\mathrm{d}x = \int \left(\frac{1}{x+2} + \frac{3x}{x^2+2}\right)\mathrm{d}x$

$= \ln|x + 2| + \frac{3}{2}\ln|x^2 + 2| + c$

$= \ln\left|(x + 2)(x^2 + 2)^{\frac{3}{2}}\right|$

When $y = 0$, $4x^2 + 6x + 2 = 0$ so $x = -1$ or $x = -\frac{1}{2} \Rightarrow x = -\frac{1}{2}$

$\therefore \int_{-\frac{1}{2}}^{0} \frac{4x^2+6x+2}{(x+2)(x^2+2)}\,\mathrm{d}x = \left[\ln\left|(x + 2)(x^2 + 2)^{\frac{3}{2}}\right|\right]_{-\frac{1}{2}}^{0}$

$= \left(\ln\left|4\sqrt{2}\right|\right) - \left(\ln\left|\frac{81}{16}\right|\right)$

$= \ln\left(\frac{16\times4\sqrt{2}}{81}\right) = \ln\left(\frac{64\sqrt{2}}{81}\right)$

**10)**

$\frac{(k-1)x^2-3kx-4}{(x-3)(x^2+4)} \equiv \frac{A}{x-3} + \frac{Bx+C}{x^2+4}$

$(k-1)x^2 - 3kx - 4 = A(x^2+4) + (Bx+C)(x-3)$

Let $x = 0$: $\quad -4 = 4A - 3C$

Compare linear: $\quad -3k = C - 3B$

Compare quadratic: $\quad k - 1 = A + B$

$\therefore A = -1, B = k$ and $C = 0$

$\frac{(k-1)x^2-3kx-4}{(x-3)(x^2+4)} = -\frac{1}{x-3} + \frac{kx}{x^2+4}$

So that $\int \frac{(k-1)x^2-3kx-4}{(x-3)(x^2+4)}\,\mathrm{d}x = \int \left(\frac{kx}{x^2+4} - \frac{1}{x-3}\right)\mathrm{d}x$

$= \frac{k}{2}\ln|x^2+4| - \ln|x-3| + c$

$= \ln\left|\frac{(x^2+4)^{\frac{k}{2}}}{x-3}\right| + c$

$\therefore \int_4^6 \frac{(k-1)x^2-3kx-4}{(x-3)(x^2+4)}\mathrm{d}x = \left[\ln\left|\frac{(x^2+4)^{\frac{k}{2}}}{x-3}\right|\right]_4^6$

$= \left(\ln\left|\frac{40^{\frac{k}{2}}}{3}\right|\right) - \left(\ln\left|\frac{20^{\frac{k}{2}}}{1}\right|\right)$

$= \ln\left(\frac{40^{\frac{k}{2}}}{3} \times \frac{1}{20^{\frac{k}{2}}}\right)$

Compare to the given $\ln\left(\frac{32}{3}\right)$

$\frac{40^{\frac{k}{2}}}{20^{\frac{k}{2}}} = 32 \Rightarrow \left(\frac{40}{20}\right)^{\frac{k}{2}} = 32 \Rightarrow 2^k = 1024$

$\therefore k = 10$